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Studierichting: Informatica
Oefeningenreeks 3H nr. 7.8

$r = \sin\theta + \cos\theta$

$\mathrm{dom} = \mathbb{R}$

$p = 2\pi$

$\to \mathrm{BD} = [0, 2\pi]$

$r = 0 \Leftrightarrow \sin\theta + \cos\theta = 0 \Leftrightarrow \sin\theta = -\cos\theta \Leftrightarrow \theta = \dfrac{3\pi}{4} + k\pi, \forall k \in \mathbb{Z}$

$\to \left\{\dfrac{3\pi}{4}, \dfrac{7\pi}{4}\right\}$

$r' = 0 \Leftrightarrow \cos\theta - \sin\theta = 0 \Leftrightarrow \cos\theta = \sin\theta \Leftrightarrow \theta = \dfrac{\pi}{4} + k\pi, \forall k \in \mathbb{Z}$

$\to \left\{\dfrac{\pi}{4}, \dfrac{5\pi}{4}\right\}$

θ	0		$\frac{\pi}{4}$		$\frac{3\pi}{4}$		$\frac{5\pi}{4}$		$\frac{7\pi}{4}$		2π
r	$+$	$+$	$+$	$+$	0	$-$	$-$	$-$	0	$+$	$+$
r'	$+$	$+$	0	$-$	$-$	$-$	0	$+$	$+$	$+$	$+$
r	$\nearrow$	$\nearrow$	M $\frac{2}{\sqrt{2}}$	$\searrow$	$\searrow$	$\searrow$	m $-\frac{2}{\sqrt{2}}$	$\nearrow$	$\nearrow$	$\nearrow$	$\nearrow$

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References